58 Prevention and Treatment of Relapse by Drugs

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58.1 Introduction

Relapse has become the most frequent cause of treatment failure after HCT (Horowitz et al. 2018). Because outcome after relapse remains poor, major efforts are focused on prevention of relapse. Beside adoptive cell-based options, such as DLI and CAR T cells, the availability of novel effective pharmacological compounds has opened new avenues in clinical research to use those drugs early after HCT in order to prevent and treat relapse (Kroger et al. 2014). The optimal pharmacological compound should have a safe toxicity profile, an antitumor effect to the underlying disease, and an immune profile which can be used to booster the graft-versus-leukemia (GVL) effect and to reduce the risk of GVHD.

58.2 Tyrosine Kinase Inhibitors (TKI) Targeting BCR/ABL

Beside a direct antitumor effect, TKIs targeting BCR/ABL in BCR/ABL-positive acute lymphoblastic leukemia or chronic myeloid leukemia are considered to induce also immunomodulating effects by inducing effect on T-cell cytolytic function, reducing T-cell PD-1 expression, and reducing myeloid-derived suppressor cells. TKIs targeting BCR/ABL such as *imatinib* induce more than 60% molecular remission in CML patients who relapsed after allograft. Smaller studies have investigated second-generation TKI successfully as maintenance therapy after allo-HCT for CML (Olavarria et al. 2007) and maintenance with TKIs after transplantation for blastic phase in CML seems to improve outcome (Niederwieser et al. 2021).

TKIs as maintenance therapy for Ph + ALL led to nonconclusive results. The CIBMTR did not find a difference in Ph + ALL patients who received posttransplant TKIs regarding relapse at 3 years, while in an EBMT study Ph + ALL patients who received TKIs posttransplant had lower relapse incidence and an improved LFS. In a small randomized study comparing TKI prophylactically or preemptive in Ph + ALL, no difference in survival was observed (Pfeifer et al. 2013). In a position statement, EBMT recommended in MRD-negative patients after allo-HCT either prophylactic or preemptive treatment (Giebel et al. 2016).

58.3 TKI Targeting FLT3-ITD

TKIs in the setting of FLT3-ITD-positive AML are of clinical relevance because a higher risk of relapse has been described for FLT3-ITD-positive patients who received allo-HCT CR1 (30% vs. 16%). Animal experiences had shown that

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sorafenib stimulated immunogenicity by induction of IL-15 which enhanced T-cell activation and GVL effect (Mathew et al. 2018a).

Midostaurin which is approved in the treatment of FLT3-positive AML has been tested in a phase II study as maintenance therapy in FLT-3-ITD-positive patients with a low relapse rate at 12 months of only 9.2%, but a randomized trial could not find a survival benefit (Maziarz et al. 2018). In contrast, two prospective randomized trials compared sorafenib to placebo as maintenance therapy after allogeneic HCT and both studies showed significant benefit for sorafenib regarding event-free and overall survival (Burchert et al. 2020; Xuan et al. 2020). A meta-analysis suggested clear benefit of TKI post-allograft in FLT3-positive AML patients, favoring sorafenib (Gagelmann et al. 2021). Currently under investigation are randomized trials with other TKIs in FLT3-positive AML such as *quizartinib*, *gilteritinib*, and *crenolanib*.

For relapsed FLT3-ITD-positive patients, sorafenib can induce long-lasting CR, and retrospective data show better outcome of sorafenib plus DLI in comparison to DLI alone (Mathew et al. 2018a; Metzelder et al. 2012).

58.4 Checkpoint Inhibitors

Checkpoint inhibitors blocking CTLA-4 and PD-1 are now widely used in solid tumors and also in hematological malignancies such as Hodgkin's disease (Ansell et al. 2015). Because of reversal of T-cell exhaustion by checkpoint inhibitors which may enhance a graft-versus-malignancy effect, this compound has also raised interest to be investigated after HCT. After auto-HCT PD-1 antibody, *pidilizumab* as maintenance therapy in DLBCL was well tolerated in a phase II study, and *nivolumab* has shown high response rate in patients with HL who relapsed after auto-HCT (Younes et al. 2016), while the combination of brentuximab vedotin plus nivolumab as consolidation strategy after auto-HCT was highly active for patients with high-risk relapsed or refractory classic HL (Herrera et al. 2023).

There is a concern about a higher risk of GVHD after checkpoint inhibition post-allograft, but *ipilimumab* did not induce high incidence of GVHD in phase I and phase II trials although the efficacy was limited with an overall response rate of less than 30% (Davids et al. 2016). PD-1 blockade investigated in a European trial was reported for 20 patients with HL who relapsed after allograft. The remission rate was high with 95% and 30% developed GVHD which was fatal in one patient. In a similar trial including 31 lymphoma patients who relapsed after allograft, the response rate was 77%, but 54% developed acute GVHD, and 8 patients died from GVHD-related complications (Haverkos et al. 2017).

58.5 Hypomethylating Agents

Methylation has a crucial role in epigenetic regulation of gene expression and malignant cells using hypermethylation to switch off a variety of genes which are responsible for growth inhibition and apoptosis. DNA methyltransferase inhibitors such as *azacytidine* or *decitabine* are active in MDS and AML, and according to their toxicity profile, they can be used after allo-HCT. Beside their effect on gene modification for differentiation and cell growth, hypomethylating agents (HMA) lead also to an upregulation of HLA and tumor-associated antigen which may be targeted by donor T cells (Hambach et al. 2009; Goodyear et al. 2010). Furthermore, CD4 and CD8 T cells were strongly suppressed by HMA, while an increase of regulatory T cells has been described.

Azacytidine and decitabine either as single agent or in combination with DLI for relapsed patients have been reported, and up to 28% CR could be achieved including long-lasting remission (Schroeder et al. 2013). In a large EBMT study, an ORR of 25% with 15% CR and a 2-year OS of 12% have been reported for azacytidine after allo-HCT relapse in AML/MDS patients. Overall, the incidence of acute GVHD was low, and the addition of DLI did not improve response or OS. Smaller studies also reported efficacy of azacytidine to convert decreasing donor cell

chimerism into full donor cell chimerism (Platzbecker et al. 2012).

Treating patients with HMA prophylactically to prevent relapse has been tested in a randomized trial, but no benefit for EFS and OS could be observed (Oran et al. 2020), and currently oral azacytidine (CC-486) is investigated in prospective randomized clinical trials.

In contrast, G-CSF in combination with decitabine has significantly reduced relapse after allogeneic HCT in a randomized trial (Gao et al. 2020).

58.6 Immunomodulating Drugs (IMiDs)

After auto-HCT, *thalidomide* has been tested alone and with glucocorticoids as maintenance to prevent relapse/progression. Most of these phase III trials demonstrated an improved PFS or EFS with variable improvement in OS, but due to toxicity, the drug has not become a standard care of treatment (Barlogie et al. 2008; Spencer et al. 2009). *Lenalidomide* is approved as maintenance therapy since a significant improvement in PFS has been shown in two randomized trials and improved OS on one randomized trial (McCarthy et al. 2012; Attal et al. 2012). A meta-analysis with data from three large studies (CALGB 100104, IFM-05-02, and GIMEMA RV-MM-PI-209) demonstrated an OS and a PFS benefit for lenalidomide maintenance. However, an increased risk of secondary primary malignancies was observed after lenalidomide maintenance therapy.

After allo-HCT, a stimulation of T cells has been shown for thalidomide, but second-generation IMiDs such as lenalidomide and *pomalidomide* induce an even more potent stimulation of T-cell-mediated immunity. IMiDs also stimulate the innate immune system including γ/δ-T cells and NK T cells. Of note, it has been shown that thalidomide even when combined with DLI was not associated with increased risk for GVHD risk (Kroger et al. 2004). Because of the stronger T-cell stimulation, lenalidomide given early post-allo-HCT can cause severe GVHD (Sockel et al. 2012), but starting with a low dose of only 5 mg and given the drug after discontinuation of IS reduces the risk of GVHD markedly (Wolschke et al. 2013).

Overall, IMiDs are potent agents for preventing relapse after auto-HCT, but their use post-allo-HCT remains to be defined primarily due to the increased risk of GvHD.

58.7 Proteasome Inhibitors

Proteasome inhibitors are mainly used as induction therapy prior auto-HCT. Some studies investigated proteasome inhibitors as maintenance therapy after auto-HCT to reduce the risk of relapse. In a prospective study, *bortezomib* as maintenance therapy was superior to thalidomide particularly in patients with renal insufficiency and high-risk cytogenetics t(4;14) or del(17q) (Goldschmidt et al. 2018).

Bortezomib after allo-HCT was tested so far only in smaller studies with acceptable rates of GVHD (Caballero-Velazquez et al. 2013), and novel proteasome inhibitors such as *ixazomib* have been investigated as maintenance therapy after allografting in MM without significant benefits (Bashir et al. 2023).

58.8 Monoclonal Antibodies

Most studies of maintenance therapy with MoAb have been conducted after auto-HCT. While maintenance therapy after autograft with anti-CD20 antibody *rituximab* failed to demonstrate an advantage for DLBCL with respect to RFS and OS (Gisselbrecht et al. 2012) for follicular lymphoma, an improved PFS but not an improvement in OS has been reported in a randomized study (Pettengell et al. 2013). An improved PFS and OS with rituximab as maintenance therapy have recently been shown for mantle cell lymphoma after auto-HCT (Le Gouill et al. 2017).

After allo-HCT for DLCBL, rituximab maintenance therapy did not improve overall survival (Glass et al. 2014). Anti-CD30 antibody drugs conjugate *brentuximab vedotin* as maintenance

therapy after auto-HCT for HL did improve PFS but not OS (Moskowitz et al. 2015).

Anti-CD22-conjugated antibody *inotuzumab ozogamicin* has been approved for relapsed ALL and has shown also activity in patients with ALL who relapsed after HCT (Kantarjian et al. 2016), but the risk of SOS/VOD is about 11% and up to 22% for those who underwent allo-HCT after inotuzumab ozogamicin.

Bispecific antibodies such as CD19-directed CD3 T-cell-engaged *blinatumomab* are active in relapsed and refractory ALL and also in MRD-positive ALL and have been investigated successfully in combination with DLI after relapse post-allo-HCT (Ueda et al. 2016). Using blinatumomab as maintenance post-allogeneic HCT was found to be feasible, while its benefit may be dependent on the immune milieu at the time of treatment (Gaballa et al. 2022).

58.9 Histone Deacetylase Inhibition (HDACI)

Histone deacetylation is a crucial mechanism of epigenetic modulation, and HDACI promotes gene expression by unwinding of histone-bound DNA. Since HDACI reduces inflammatory cytokines and increases T-regulatory cells, the drug was also used for GVHD prevention in a phase I/II study (Choi et al. 2014). *Panobinostat* was tested in two trials as maintenance therapy after allo-HCT in AML/MDS with or without (Bug et al. 2017) DLI resulting in an encouraging 1-year RFS of 66% in combination with DLI and 2-year RFS of 74% if used as single agent.

58.10 BCL-2 Inhibitors

BCL-2 inhibitor (venetoclax) has shown promising results in treatment of acute myeloid leukemia. Small phase II studies using *venetoclax plus decitabine* to prevent relapse have been reported, but until results from randomized phase III are available the drugs should be used only in clinical trials.

58.11 IDH1/IDH2 Inhibitors

IDH1 and IDH2 inhibitors ivosidenib and enasidenib have shown efficacy and favorable safety profile in the treatment of IDH1- or IDH2-positive acute leukemia and also in early phase I studies as maintenance post-allograft (Fathi et al. 2023).

Drug	Disease	Setting	Impact
Imatinib	CML	Prophylactic	Beneficial
	Ph + ALL		No benefit
Nilotinib	Ph + ALL	Prophylactic/preemptive	Unclear
Dasatinib	Ph + ALL	Prophylactic	Unclear
Sorafenib	FLT3-ITD AML	Maintenance	Beneficial
Midostaurin	FLT3-ITD AML	Maintenance	May be beneficial
Gilteritinib	FLT3-ITD AML	Maintenance	No benefit
Pidilizumab	DLBCL	Prophylactic	Unclear
Nivolumab	HL	Consolidation	Beneficial
Pembrolizumab	HL	Consolidation	Beneficial
Ipilimumab	Myeloid+lymphoid	Treatment	Unclear
Azacytidine (+DLI)	AML/MDS	Treatment	May be beneficial
		Prophylactic	No benefit
Decitabine (+DLI)	AML/MDS	Treatment	May be beneficial
Decitabine + GCSF	AML	Prophylactic	Beneficial
Ivosidenib	IDH1-mutated AML	Maintenance	Unclear
Enasidenib	IDH2-mutated AML	Maintenance	Unclear

Drug	Disease	Setting	Impact
IMiDs (Len, Thal, Pom)	Myeloma	Maintenance post-auto	Standard
		Post-allo	Unclear
PI (Bor, Car, Ixa)	Myeloma	Maintenance post-auto	May be beneficial in HR
		Post-allo	Unclear
Brentuximab vedotin	HL	Maintenance post-auto	Beneficial
		Post-allo	Unclear
Rituximab	DLBCL	Maintenance post-auto	No benefit
		Post-allo	
	MCL	Maintenance post-auto	Beneficial
		Post-allo	Unclear
Inotuzumab ozogamicin	ALL	Treatment	Unclear
Blinatumomab (+DLI)	ALL	Treatment	Beneficial
		Maintenance	May be beneficial
Venetoclax (+DLI/AZA)	AML	Prophylactic	Unclear

AML acute myeloid leukemia, *MDS* myelodysplastic syndromes, *ALL* acute lymphoblastic leukemia, *Ph+* Philadelphia chromosome-positive, *HL* Hodgkin lymphoma, *DLBCL* diffuse large B-cell lymphoma, *MCL* mantle cell lymphoma, *DLI* donor lymphocyte infusion, *Len* lenalidomide, *Thal* thalidomide, *Pom* pomalidomide, *Bor* bortezomib, *Car* carfilzomib, *Ixa* ixazomib, *AZA* azacytidine, *HR* high risk

Key Points

- Outcome after relapse to allogeneic stem cells remains poor, and major efforts should focus on prevention of relapse.
- Beside adoptive cell-based options such as DLI, the availability of novel effective pharmacological compounds has opened new avenues in clinical research, mainly:
  - Tyrosine kinase inhibitors (TKI) targeting BCR/ABL
  - TKI targeting FLT3-ITD (sorafenib, midostaurin, quizartinib, gilteritinib, crenolanib)
  - Checkpoint inhibitors (pidilizumab, nivolumab, ipilimumab, pembrolizumab)
  - Hypomethylating agents (azacytidine, decitabine)
  - Immunomodulating drugs (thalidomide, lenalidomide, pomalidomide)
  - Proteasome inhibitors (bortezomib, ixazomib)
  - Antibodies (rituximab, brentuximab vedotin, inotuzumab ozogamicin, blinatumomab)
  - Histone deacetylase inhibition (panobinostat)
  - BCL-2 inhibitor (venetoclax)
  - IDH1/IDH2 inhibitors (ivosidenib and enasidenib)
- The optimal pharmacological compound should have a safe toxicity profile, an antitumor effect to the underlying disease, and an immune profile which can be used to booster the GVL effect and to reduce the risk of GVHD.

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